Shiliang SHAO (邵世梁)

No. 37 Xueyuan Road Haidian District Beijing, 100191, China
Tel. +86 18296223412 Email: ssl020104@163.com
Personal Website: https://shiliangshao.github.io/

EDUCATION

Beihang University (BUAA; Project 985), Beijing, China — Sept 2024 – Jun 2027

- M.Eng. in **Mechanical Engineering**; GPA: 90.31/100 (3.80 / 4.00); Class Rank: **3/13**
- Core Modules: *Intelligent Optimization Methods, Matrix Theory and Applications, Mathematical Statistics, Science Ethics and Academic Specifications*
- Advisors: Prof. Maolin Cai, Prof. Xiaomeng Tong

Civil Aviation University of China (CAUC), Tianjin, China — Sept 2020 – Jun 2024

- B.Eng. in Electronic Information Engineering; GPA: 89.58/100; Departmental Rank: **1/375**
- Core Modules: *Advanced Mathematics A (I, II); Linear Algebra; Probability Theory and Mathematical Statistics; C Programming*
- Selected Honors: Outstanding Undergraduate **Thesis** (2024); Outstanding **Graduate** of CAUC (2024)

PUBLICATIONS

Published

- Qu, H., **Shao, S.**, Xu, Y., Cai, M., Shi, Y., & Tong, X. (2026). A global feedback learning-based memetic algorithm for energy-aware scheduling in collaborative heterogeneous flexible job shops. *Swarm and Evolutionary Computation*, *104*, 102392. [JCR Q1, IF = 9.6]

In Revision

- **Shao, S.**, Cai, M., Qu, H., Tong, X., & Gao, X. Offline reinforcement learning with heterogeneous graph representation and attention mechanisms for human–machine collaborative flexible job shop scheduling. *Neurocomputing*. [JCR Q1, IF = 6.7]

Conference

- **Shao, S.**, Xu, Y., Qu, H., Cai, M., Hu, Y., & Tong, X. (2025, September). AVS-ESA: A multi-objective optimization approach for energy-efficient variable-speed flexible job shop scheduling. In *2025 China Automation Congress (CAC)* (pp. 6750-6755). IEEE.

Under Review

- Qu, H., Ji, J., Xu, S., **Shao, S.**, Sun, X., Han, C., Sun, Z., & Shi, Y. Bias-guided policy search for human–robot collaborative flexible job shop scheduling via deep reinforcement learning. *Journal of Systems Engineering and Electronics*. [JCR Q2, IF = 2.6]

RESEARCH EXPERIENCE

Graph Attention & Offline Reinforcement Learning for Flexible Job-Shop Scheduling — Sept 2025 – Jun 2027

M.Eng. Thesis Research — Advisors: Prof. M. Cai, Prof. X. Tong

- Tackled factory scheduling when **workers**, not just machines, must be assigned, building a model that decides **which job runs on which machine with which worker** to finish the workload as **early** as possible
- Developed MA-DQUAC, a method that learns from **existing data instead** of repeated trial runs on a real production line, combining a graph network, an attention mechanism, and a technique that keeps the model from becoming overconfident on unfamiliar cases
- **Tested** it against standard approaches including dispatching rules, genetic algorithms, OR-Tools, and existing reinforcement learning, where it produced better schedules, needed less training data, and stayed **reliable** as problems grew larger

INDUSTRY EXPERIENCE

Aerfeisi (Jiangsu) Robotics Technology Co., Ltd. — Jul 2025 – Sept 2025

Algorithm Engineer (Intern) — *Wuxi, China*

- Worked on the chassis control of a mobile robot, computing wheel speeds and steering angles across different drive modes and improving the state-machine switching between them so the robot moved more smoothly and safely (Linux / ROS / C++)
- Helped build a vision-based autonomous navigation pipeline, training a segmentation model to identify drivable areas and obstacles and wiring it into the ROS perception and control modules so the robot could avoid obstacles on its own

EXTRACURRICULARS

- **Executive Director**, Liaison and Exchange Center, Beihang Graduate Student Union: Organized inter-university academic exchange events and coordinated graduate-student liaison activities across the school. Dec 2024 – Dec 2025
- **President**, Innovation and Entrepreneurship Center, Civil Aviation University of China: Led a 30+ member student organization, coordinating university-wide competition teams and undergraduate research-training initiatives. Dec 2022 – Dec 2023
- **Participant**, Summer Academic Exchange Program, The Hong Kong Polytechnic University & Hong Kong International Aviation Academy: Completed a two-week program on aviation engineering and intelligent manufacturing. Jun 2023 – Jul 2023

PATENTS

- Tong, X., Cai, M., **Shao, S.**, & Qu, H. 一种基于离线强化学习的人机协作柔性作业车间调度方法 A method for human–machine collaborative flexible job shop scheduling based on offline reinforcement learning. Chinese invention patent application, CN122155265A.
- Tong, X., Cai, M., **Shao, S.**, & Fang, M. 一种基于离线强化学习并结合在线微调方法的双臂机器人 A dual-arm robot control method based on offline reinforcement learning and online fine-tuning. Chinese invention patent application, CN122110724A.
- Tong, X., Liu, Q., **Shao, S.**, & Cai, M. 一种基于深度神经模糊推理系统 A method based on a deep neuro-fuzzy inference system. Chinese invention patent application, CN121413450A.
- Cai, M., Tong, X., Qu, H., Ning, F., & **Shao, S.** 多策略柔性作业车间调度方法 A multi-strategy energy-efficient flexible job shop scheduling method. Chinese invention patent application, CN119847094A.

SOFTWARE COPYRIGHTS

- Tong, X., Qu, H., **Shao, S.**, & Cai, M. An LLM-driven intelligent scheduling algorithm generation system for heterogeneous flexible job shops. Software copyright registration, No. 2026SR0249508, China, 2026.
- Tong, X., **Shao, S.**, & Cai, M. An offline reinforcement learning-driven human–machine collaborative flexible job shop scheduling system. Software copyright registration, No. 2026SR0556778, China, 2026.

SELECTED HONORS & AWARDS

Academic Honors

- First-Class Scholarship: Beihang University (2025); CAUC (2021–2023)
- Outstanding Student Leader: Beihang University (2025); CAUC (2021–2023)
- SMC Corporation Scholarship (2026) and Hisense Scholarship (2025; sole recipient in the college)
- **National Scholarship**, Ministry of Education of China (2021, 2022, 2023)
- "Top Ten Outstanding Undergraduates," CAUC (2024)
- Outstanding Undergraduate of Tianjin Municipality (2023; sole awardee among 375 peers)
- Innovation & Entrepreneurship Scholarship, CAUC (2022, 2023; ~10 per college annually)

Competition Awards

- Second Prize (**National**), China Undergraduate Mathematical Contest in Modeling — Team **Captain** (2022)
- Successful Participant, Mathematical Contest in Modeling (MCM/ICM) — Team Captain (2022)
- **National** Undergraduate Electronics Design Contest, Tianjin Division — Team Captain: Second Prize (2022), Third Prize (2021, 2023)
- China International "Internet+" Innovation & Entrepreneurship Competition, Tianjin: **Gold** (7^{th}, 2021), Silver (7^{th}, 2021; 8^{th}, 2022)

- Second Prize, 13th BeiDou-Cup National Youth Science & Technology Innovation Contest, Tianjin Division — Paper Track (2022)
- Second Prize, North China Five-Province Robotics Competition, Tianjin Division — Team Captain (2021)

SKILLSET

- **Languages:** English (IELTS 6.5; June 2026), Mandarin
- **Programming Languages**: Python (primary), C, C++, MATLAB
- **ML & Optimization**: PyTorch, scikit-learn, NumPy, SciPy, Pandas, OR-Tools
- **Engineering & CAE**: SolidWorks, Altair HyperMesh, Altair Radioss, Simulink, Altium Designer, Multisim
- **Tools & Environment**: Linux (Ubuntu), Git, ROS, LaTeX